

A Matrix-Degree Obstruction to Rational Generation of Boolean-Lattice Pseudo-Roots

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Abstract

For the neighborhood seed associated with the four-vertex path P_4 , the diamond operations do not recover all Boolean-lattice pseudo-roots. The corresponding question for unrestricted rational operations in the free skew field is subtler: the seed map has an invertible linearization and therefore a unique formal inverse near every generic scalar point. We prove that this formal inverse is not free rational. A symmetric one-parameter curve of 2×2 matrix outputs has a formal inverse whose coefficient field contains an element of degree three over $\mathbb{Q}(t)$. An exact elimination in a quadratic Pauli algebra produces the irreducible cubic. Its conjugate inverse branches are unramified, forcing the generic matrix degree of the seed map to be at least three in every size $n \geq 2$. This contradicts the degree-one consequence of any free rational inverse. The same matrix-degree argument, without specializing a hypothetical inverse, extends the obstruction to every graph containing an induced P_4 .

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1 Introduction

Let B_n be the Boolean lattice on $[n] = \{1, \dots, n\}$. We draw its cover edges in the inclusion direction, so the edge $A \rightarrow A \cup \{i\}$ is labelled by a pseudo-root $x_{A,i}$. The Boolean-lattice model and its diamond relations arise from factorizations of noncommutative polynomials; see [3, 4, 9]. If two lower pseudo-roots are denoted by x and y , the upper pseudo-roots in the corresponding diamond are $C(y, x)$ and $C(x, y)$, where

$$C(y, x) = (y - x)^{-1}x(y - x), \quad D(y, x) = (y - x)x(y - x)^{-1}. \quad (1)$$

Thus C denotes the lower-to-upper operation and D its inverse. In the terminology of Retakh and Saks, these are the D - and U -operations, respectively. All free rational expressions below are interpreted in the free skew field, or on matrix tuples for which every displayed inverse exists. We use the standard universal-field and matrix-evaluation framework for noncommutative rational functions [1, 6].

For a graph G on $[n]$, its neighborhood seed consists of the n pseudo-roots

$$Y_G = \{x_{N_G(i),i} : i \in [n]\}.$$

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At the level of diamond closure, the cographs are precisely the graphs for which Y_G generates all pseudo-roots [8, Theorem 1.1]. Cographs are equivalently the finite graphs with no induced P_4 [2]. The first graph outside this class is the path P_4 . Write its bottom pseudo-roots as a, b, c, d . Its seed map is

$$F_{P_4}(a, b, c, d) = (C(b, a), C(C(a, c), C(a, b)), C(C(b, d), C(b, c)), C(c, d)). \quad (2)$$

The main result is the following.

Theorem 1.1. *Let K be a field of characteristic zero. If*

$$\begin{aligned} p &= C(b, a), & q &= C(C(a, c), C(a, b)), \\ r &= C(C(b, d), C(b, c)), & s &= C(c, d), \end{aligned}$$

then

$$K\langle p, q, r, s \rangle_{\text{rat}} \subsetneq K\langle a, b, c, d \rangle_{\text{rat}}.$$

Equivalently, F_{P_4} has no free rational inverse.

The proof uses matrix evaluations, but not merely a failure at one fixed matrix size. For every $n \geq 2$, the rational map induced by F_{P_4} on four $n \times n$ matrices has generic degree at least three. A finite collection of free rational recovery expressions necessarily has a common matrix domain at some size, where it would give a rational left inverse and force generic degree one. This all-size argument avoids the polynomial-identity loophole inherent in a single small-matrix specialization.

The only computer-assisted step is an exact resultant calculation. The ancillary file `p4_matrix_degree_certificate.py` performs the calculation over $\mathbb{Q}$, verifies the formal branch used to select the relevant factors, and checks the irreducibility specialization. No floating-point arithmetic or numerical root selection is used.

2 Diamond inversion and the P_4 reductions

Lemma 2.1 (Diamond inversion). *Let*

$$u = C(y, x), \quad v = C(x, y).$$

Then

$$v - u = y - x, \quad x = D(v, u), \quad y = D(u, v).$$

Consequently, the rational transformations

$$(x, y) \mapsto (C(y, x), C(x, y)) \quad \text{and} \quad (u, v) \mapsto (D(v, u), D(u, v))$$

are mutually inverse wherever their displayed inverses exist.

Proof. Since $x - y = -(y - x)$,

$$v = (y - x)^{-1}y(y - x), \quad u = (y - x)^{-1}x(y - x).$$

Thus $v - u = y - x$. It follows that

$$D(v, u) = (v - u)u(v - u)^{-1} = (y - x)(y - x)^{-1}x(y - x)(y - x)^{-1} = x.$$

The proof of $D(u, v) = y$ is identical. Substituting these formulas back into C proves the final assertion. $\square$

For $P_4 = 1 - 2 - 3 - 4$, set

$$a = x_{\emptyset,1}, \quad b = x_{\emptyset,2}, \quad c = x_{\emptyset,3}, \quad d = x_{\emptyset,4}.$$

The four neighborhood seeds are

$$\begin{aligned} p &= x_{\{2\},1} = C(b, a), \\ q &= x_{\{1,3\},2} = C(C(a, c), C(a, b)), \\ r &= x_{\{2,4\},3} = C(C(b, d), C(b, c)), \\ s &= x_{\{3\},4} = C(c, d). \end{aligned} \tag{3}$$

Define the missing adjacent partners

$$T = C(a, b), \quad W = C(d, c). \tag{4}$$

Proposition 2.2 (Partner reduction). *The map F_{P_4} is free rationally invertible if and only if T and W are simultaneously free rational functions of p, q, r, s .*

Proof. If T and W are recoverable, Lemma 2.1 gives

$$a = D(T, p), \quad b = D(p, T), \quad d = D(W, s), \quad c = D(s, W).$$

Conversely, recovery of a, b, c, d gives $T = C(a, b)$ and $W = C(d, c)$. $\square$

There is also a useful conjugator form of this reduction. Put

$$u = b - a, \quad v = c - d.$$

The relation in Lemma 2.1 gives

$$T - p = u, \quad W - s = v, \quad a = upu^{-1}, \quad b = upu^{-1} + u, \quad d = vsv^{-1}, \quad c = vsv^{-1} + v. \tag{5}$$

Hence

$$K\langle a, b, c, d \rangle_{\text{rat}} = K\langle p, s, u, v \rangle_{\text{rat}},$$

and partner recovery is equivalent to recovery of u and v .

For completeness, define the second-layer partners

$$M = x_{\{1,2\},3}, \quad N = x_{\{3,4\},2}.$$

The relevant diamonds give

$$T = D(M, q), \quad W = D(N, r),$$

so Proposition 2.2 is equivalently a recovery problem for M, N . An exact two-variable consistency system is

$$D(q, M) = C(D(D(M, q), p), D(s, D(N, r))), \tag{6}$$

$$D(r, N) = C(D(D(N, r), s), D(p, D(M, q))). \tag{7}$$

For the true partners, the left sides are respectively $C(a, c)$ and $C(d, b)$, so the equations hold. Conversely, suppose M, N solve (6)–(7). Define

$$T = D(M, q), \quad W = D(N, r), \quad a = D(T, p), \quad b = D(p, T), \quad d = D(W, s), \quad c = D(s, W).$$

The adjacent pairs reconstruct p, s by Lemma 2.1. The two consistency equations say

$$D(q, M) = C(a, c), \quad D(r, N) = C(d, b).$$

Applying the mutually inverse diamond transformations to the upper pairs (q, M) and (r, N) now reconstructs the middle seeds q, r . Thus every solution for which the displayed inverses exist gives a preimage of (p, q, r, s) .

3 Formal invertibility and matrix degree

We record the distinction between formal and rational inversion. Let $\lambda_1, \dots, \lambda_g$ be distinct central scalars and write a matrix or free variable as $x_i = \lambda_i + X_i$, where X_i has positive degree.

Lemma 3.1 (First-order conjugation). *If $x = \lambda + X$, $y = \mu + Y$, and $\lambda \neq \mu$ are central scalars, then*

$$C(y, x) = \lambda + X + O(2), \quad D(y, x) = \lambda + X + O(2).$$

Proof. Put $\delta = \mu - \lambda$ and $E = Y - X$. Then

$$(y - x)^{-1} = (\delta + E)^{-1} = \delta^{-1} - \delta^{-2}E + O(2).$$

Substitution into $(y - x)^{-1}x(y - x)$ shows that the degree-zero and degree-one terms are λ and X ; the terms involving λE cancel because λ and δ are central. The calculation for D is the same. $\square$

Proposition 3.2 (Formal inverse). *At every scalar point*

$$(a, b, c, d) = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$$

with distinct λ_i , the linear part of F_{P_4} is the identity. Consequently, F_{P_4} has a unique inverse in the completed noncommutative power-series algebra at that point. For every matrix size n , the induced rational map $F_{P_4, n}$ has identity Jacobian there and is dominant and generically finite.

Proof. Every use of C retains, to first order, the perturbation of its second argument by Lemma 3.1. Reading the four expressions in (3) from the inside out gives

$$p = \lambda_1 + A + O(2), \quad q = \lambda_2 + B + O(2), \quad r = \lambda_3 + G + O(2), \quad s = \lambda_4 + J + O(2).$$

An endomorphism of a complete filtered algebra with invertible linear part has a unique formal inverse, obtained recursively by degree. The same linearization on $n \times n$ matrix entries is the identity map on an affine space of dimension $4n^2$. A rational self-map with a nonsingular Jacobian at one point is dominant; equal source and target dimensions then imply generic finiteness. $\square$

We shall use two elementary degree principles. Degrees are geometric degrees, computed after extension to an algebraic closure. Our conventions for rational maps, function fields, and birationality are standard; see [5, Chapter I, §4].

Lemma 3.3 (Simple points bound the generic degree). *Let $f : X \dashrightarrow Y$ be a dominant generically finite rational map between irreducible smooth varieties of the same dimension over a characteristic-zero field. If one geometric fiber contains r distinct points in the domain of f at which the Jacobian determinant is nonzero, then*

$$\deg(f) \geq r.$$

Proof. All data involved in the points and the map are defined over a finitely generated characteristic-zero field, which embeds in $\mathbb{C}$. After this embedding, the complex inverse function theorem gives pairwise disjoint analytic neighborhoods of the r points and a common analytic neighborhood of their image on which each neighborhood supplies a distinct local inverse branch. A dominant generically finite map restricts, after deleting proper algebraic subsets of source and target, to a finite étale map of degree $\deg(f)$. The deleted subsets and their inverse images under the local branches have empty analytic interior. A nearby point can therefore be chosen in the finite étale locus and on every local branch. Its fiber has at least r points but exactly $\deg(f)$ points. Hence $\deg(f) \geq r$. $\square$

Lemma 3.4 (Free left inverses force matrix degree one). *Let F be a free rational self-map in g variables. Suppose every matrix evaluation F_n is dominant. If F has a free rational left inverse, then F_n is birational for at least one $n \geq 2$.*

Proof. Choose rational expressions for the finitely many components of the proposed left inverse. Each expression has a nonempty matrix domain at some size. Domains are stable under direct sums, so after taking a common multiple of these sizes, enlarged if necessary to be at least two, every component has a nonempty domain at one common size n . At fixed size, the domain of a rational expression is Zariski open. Since the matrix tuple space is irreducible, the finitely many nonempty domains have nonempty intersection.

Thus the left inverse determines a rational map R_n on a nonempty open subset of the target matrix space. Dominance of F_n ensures that the composition $R_n \circ F_n$ is defined on a nonempty open subset. Equality of the free rational functions gives

$$R_n \circ F_n = \text{id}$$

as rational maps. On function fields this becomes

$$F_n^* \circ R_n^* = \text{id}.$$

The injective homomorphism F_n^* is therefore also surjective, so it is an isomorphism. Hence F_n is birational. $\square$

4 A symmetric 2×2 output curve

Let

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (8)$$

They satisfy

$$X^2 = Z^2 = I, \quad Y^2 = -I,$$

and anticommute pairwise. Consider the output curve

$$p = tX, \quad q = I + tX, \quad r = 2I + tZ, \quad s = 3I + tZ. \quad (9)$$

At $t = 0$ this is the scalar point $(0, I, 2I, 3I)$. By Proposition 3.2, it has a unique formal preimage

$$a = A, \quad b = I + B, \quad c = 2I + G, \quad d = 3I + J \quad (10)$$

in $\text{Mat}_2(\mathbb{Q}[[t]])^4$.

The symmetry of this branch reduces the elimination to three scalar coordinates. Set

$$H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \kappa(M) = (HMH^{-1})^\top.$$

Since $H^\top = H$ and $H^2 = 2I$, the map κ is an anti-involution. Direct calculation gives

$$\kappa(X) = Z, \quad \kappa(Y) = Y, \quad \kappa(Z) = X. \quad (11)$$

For traceless 2×2 matrices U, V , define

$$\Gamma_\delta(V, U) = (\delta I + V - U)^{-1}U(\delta I + V - U).$$

This is the centered part of $C(\mu I + V, \lambda I + U)$ when $\delta = \mu - \lambda$.

Lemma 4.1 (Reversal identities). *For traceless 2×2 matrices,*

$$\kappa(\Gamma_\delta(V, U)) = \Gamma_{-\delta}(\kappa V, \kappa U). \quad (12)$$

Moreover, whenever the expressions are defined,

$$C(C(x, z), C(x, y)) = C(C(z, x), C(z, y)). \quad (13)$$

Proof. If R is traceless, then R^2 is scalar and $(\delta I + R)^{-1}$ is a scalar multiple of $\delta I - R$. Applying the anti-involution to the defining conjugation for Γ_δ therefore replaces its conjugator, up to an irrelevant nonzero scalar, by $-\delta I + \kappa V - \kappa U$. This proves (12).

For (13), put $A = x - y$ and $B = z - y$. The elementary identity

$$B(A - B)^{-1}A = A(A - B)^{-1}B$$

follows by substituting $A = (A - B) + B$ into both sides. Substitution into the definitions of C then gives

$$(x - y)(C(x, z) - C(x, y)) = (z - y)(C(z, x) - C(z, y)).$$

Writing the common element as h , both sides of (13) are $h^{-1}yh$. □

Let $\widehat{F}$ be the centered form of F_{P_4} , so that

$$F_{P_4}(A, I + B, 2I + G, 3I + J) = (P, I + Q, 2I + R, 3I + S).$$

On traceless tuples define

$$\tau(A, B, G, J) = (\kappa J, \kappa G, \kappa B, \kappa A).$$

Equations (12) and (13), applied from the inner diamonds outward, give

$$\widehat{F} \circ \tau = \tau \circ \widehat{F}. \quad (14)$$

The first and fourth coordinates follow directly from (12); equation (13) changes the common lower vertex in each middle coordinate before (12) is applied, interchanging the second and third coordinates.

Every centered matrix in (10) is traceless. Indeed, p, q, r, s are respectively conjugate to a, b, c, d , and the output curve has scalar parts 0, 1, 2, 3 and traceless perturbations. The centered output tuple in (9) is (tX, tX, tZ, tZ) , which is fixed by τ . Formal uniqueness and (14) therefore imply

$$G = \kappa(B), \quad J = \kappa(A). \quad (15)$$

Recursive substitution through degree three gives

$$\begin{aligned} A &= tX + 4t^3Z + O(t^4), \\ B &= tX + 2t^2Y + (2X + 4Z)t^3 + O(t^4), \\ G &= tZ + 2t^2Y + (4X + 2Z)t^3 + O(t^4), \\ J &= tZ + 4t^3X + O(t^4). \end{aligned} \quad (16)$$

These identities are also checked directly by the ancillary exact-arithmetic script.

5 Exact elimination on the formal branch

Because q is conjugate to b , equation (9) gives

$$\operatorname{tr}(B) = 0, \quad B^2 = t^2 I.$$

The matrices X, Y, Z form a basis of the traceless 2×2 matrices, so there are unique $x, y, z \in \mathbb{Q}[[t]]$ such that

$$B = xX + yY + zZ, \quad x^2 - y^2 + z^2 = t^2. \quad (17)$$

By (15),

$$G = zX + yY + xZ.$$

The jet (16) gives

$$x = t + 2t^3 + O(t^4), \quad y = 2t^2 + O(t^4), \quad z = 4t^3 + O(t^4). \quad (18)$$

The equation $p = C(b, a) = tX$ determines A rationally from B . Since A is conjugate to tX , we have $A^2 = t^2 I$. The relation $(b - a)p = a(b - a)$ then implies

$$(b + p)p - a(b + p) = p^2 - a^2 = 0.$$

Thus, with

$$L = I + B + tX, \quad e = 2t^2 + 2tx - 1,$$

we have $A = L(tX)L^{-1}$. Here $\det L = 1 - 2t^2 - 2tx = -e$, so L is invertible on the formal branch. Calculation in the Pauli basis gives

$$\begin{aligned} A = & \frac{t(2x(t+x)-1)}{e}X + \frac{2t((t+x)y-z)}{e}Y \\ & + \frac{2t((t+x)z-y)}{e}Z. \end{aligned} \quad (19)$$

Put

$$T = C(a, b), \quad U = C(a, c), \quad h = (b - a)(U - T).$$

Since $T = (a - b)^{-1}b(a - b)$ and $q = (U - T)^{-1}T(U - T)$,

$$q = h^{-1}bh.$$

Using $q = I + tX$ and $b = I + B$, this is equivalent to

$$E := htX - Bh = 0. \quad (20)$$

Let $\mathcal{A} = \mathbb{Q}(t, x, z)(y)$ be the quadratic field determined by

$$y^2 = x^2 + z^2 - t^2.$$

The radicand has simple zeros when regarded as a rational function of x over $\mathbb{Q}(t, z)$, so it is not a square in $\mathbb{Q}(t, x, z)$. Substitute (17), (19), and $G = zX + yY + xZ$ into (20), calculating with

$$XY = -Z, \quad XZ = -Y, \quad YZ = -X$$

and anticommutation. Write the scalar and X -coordinates of E as

$$E_0 = a_0 + b_0 y, \quad E_X = a_1 + b_1 y,$$

where $a_i, b_i \in \mathbb{Q}(t, x, z)$. On the branch (18), exact expansion gives

$$b_0 = -t^2 + O(t^4), \quad b_1 = t + O(t^3). \quad (21)$$

In particular, both are nonzero in $\mathbb{Q}((t))$. Therefore

$$y = \rho_0 := -\frac{a_0}{b_0} = \rho_1 := -\frac{a_1}{b_1}$$

on the formal branch.

Cancel each ρ_i in $\mathbb{Q}(t, x, z)$ before clearing denominators. Let $S(t, x, z)$ be the unique irreducible factor vanishing at $(0, 0, 0)$ in the numerator of $\rho_0 - \rho_1$, and let $N(t, x, z)$ be the corresponding factor in the numerator of

$$\rho_0^2 - (x^2 + z^2 - t^2).$$

All other irreducible factors in these two numerators have nonzero value at the origin and hence are units after substitution of the formal branch. Thus

$$S(t, x(t), z(t)) = N(t, x(t), z(t)) = 0. \quad (22)$$

Lemma 5.1 (Exact elimination). *The factors just defined have x -degrees 4 and 8, respectively, and*

$$\begin{aligned} \text{Res}_x(S, N) &= 16t^{12}z(z-t)^2(t-1)^4(t+1)^4(2t-1)^3(2t+1)^3 \\ &\quad \cdot (3t^2 + tz - 2)^2 (8t^4 + 4t^3z + 4t^2z^2 - 8t^2 - 4tz + 1)^2 P(t, z), \end{aligned} \quad (23)$$

where

$$P(t, z) = A_3(t)z^3 + A_2(t)z^2 + A_1(t)z + A_0(t)$$

and

$$\begin{aligned} A_3(t) &= 256t^{12} - 1472t^{10} + 2096t^8 - 676t^6 + 160t^4 - 25t^2, \\ A_2(t) &= 256t^{11} - 1984t^9 + 1808t^7 + 28t^5 + 86t^3 + 20t, \\ A_1(t) &= -2112t^{10} + 1552t^8 + 1860t^6 - 257t^4 + 140t^2 - 4, \\ A_0(t) &= -2880t^9 + 672t^7 - 564t^5 + 16t^3. \end{aligned}$$

Proof. This is a finite exact calculation in the quadratic algebra $\mathcal{A}$. The ancillary script constructs A, B, G, T, U, h, E , extracts S, N by exact factorization over $\mathbb{Q}$, and verifies (23) by expanding the difference between its two sides. It also verifies the nonzero leading terms in (21). Every operation is symbolic over $\mathbb{Q}$; no numerical approximation enters the calculation. $\square$

From (22), the left side of (23) vanishes after substitution of the formal branch. In the integral domain $\mathbb{Q}[[t]]$, the factors t, z , and $z - t$ are nonzero by (18); every other displayed factor except P has nonzero constant term. Hence

$$P(t, z(t)) = 0. \quad (24)$$

Lemma 5.2 (Irreducibility of the cubic). *The polynomial $P(t, z)$ is irreducible in $\mathbb{Q}(t)[z]$.*

Proof. The coefficient polynomials A_3, A_2, A_1, A_0 have greatest common divisor one, so P is primitive in $\mathbb{Q}[t][z]$. At $t = 2$,

$$P(2, z) = 4(9255z^3 - 64618z^2 - 412473z - 351616).$$

The cubic in parentheses is primitive. Modulo 7 it becomes

$$z^3 - z^2 + 2z + 1,$$

whose values at $0, 1, \dots, 6$ are $1, 3, 2, 4, 1, 6, 4$. It has no root in $\mathbb{F}_7$ and is therefore irreducible there, hence irreducible over $\mathbb{Q}$. The leading coefficient $A_3(2)$ is nonzero. If P factored over $\mathbb{Q}(t)$, Gauss's lemma would give a factorization in $\mathbb{Q}[t, z]$; specialization at $t = 2$ would preserve the positive z -degrees of both factors and would factor $P(2, z)$, a contradiction. $\square$

Combining (24) with Lemma 5.2 gives

$$[\mathbb{Q}(t, z(t)) : \mathbb{Q}(t)] = 3. \quad (25)$$

6 The matrix-degree obstruction

Let F_n be the rational map induced by F_{P_4} on four $n \times n$ matrices. By Proposition 3.2, F_n is dominant and generically finite. We now convert (25) into a lower bound for its degree.

Proposition 6.1. *For every $n \geq 2$,*

$$\deg(F_n) \geq 3.$$

Proof. First let $n = 2$. Let L be the field over $\mathbb{Q}(t)$ generated by all entries of the formal preimage (10). The equations defining the inverse branch are rational algebraic equations whose denominators have nonzero constant terms. Their Jacobian with respect to the input entries is the identity at $t = 0$. The algebraic implicit-function theorem therefore shows that every entry of the branch is algebraic over $\mathbb{Q}(t)$; equivalently, $L/\mathbb{Q}(t)$ is finite. Since $z(t) \in L$, equation (25) gives

$$[L : \mathbb{Q}(t)] \geq 3.$$

The extension is separable. Each $\mathbb{Q}(t)$ -embedding of L into an algebraic closure sends the matrix entries of the branch to another preimage of the same output curve. The resulting points are distinct because those entries generate L . Every denominator that occurs in F_2 is a nonzero element of L , so it remains nonzero under each embedding. The Jacobian determinant on the original branch is $1 + O(t)$ and hence is nonzero in L ; its images under the embeddings are also nonzero. Thus the fiber over (9) contains at least three distinct simple points. Lemma 3.3 gives $\deg(F_2) \geq 3$.

For $n > 2$, append to every input matrix the scalar block corresponding to its center:

$$A \oplus 0I_{n-2}, \quad (I + B) \oplus I_{n-2}, \quad (2I + G) \oplus 2I_{n-2}, \quad (3I + J) \oplus 3I_{n-2}.$$

The map C respects direct sums, so the conjugate 2×2 preimages produce at least three distinct preimages of one $n \times n$ block-diagonal output curve. On the original amplified branch the input tends at $t = 0$ to the scalar tuple $(0I_n, I_n, 2I_n, 3I_n)$, so its $n \times n$ Jacobian determinant is $1 + O(t)$ by Proposition 3.2. This determinant is a nonzero element of L , and its conjugates are therefore nonzero as well. The three amplified points are simple, and Lemma 3.3 gives $\deg(F_n) \geq 3$. $\square$

The maps F_n are defined over $\mathbb{Q}$, and the degree in Proposition 6.1 is geometric. It is therefore unchanged by extension to any characteristic-zero constant field.

Proof of Theorem 1.1. If a, b, c, d were free rational functions of p, q, r, s , those expressions would define a free rational left inverse of F_{P_4} . By Lemma 3.4, F_n would then be birational for some $n \geq 2$, so $\deg(F_n) = 1$. This contradicts Proposition 6.1. $\square$

Corollary 6.2. *The adjacent partners $T = C(a, b)$ and $W = C(d, c)$ are not simultaneously free rational functions of p, q, r, s . The same holds for the second-layer partners M, N .*

Proof. Apply Proposition 2.2 and the equivalent second-layer recovery described after it. $\square$

7 Extension from P_4 to non-cographs

We finish by replacing a potentially invalid specialization of a hypothetical rational inverse with a specialization of the explicit algebraic branch. The latter has all denominators controlled at its scalar base point.

For a graph G on $[m]$, let F_G denote the rational map from the m bottom pseudo-roots to the neighborhood seed Y_G . Its evaluation on $n \times n$ matrices is denoted $F_{G,n}$.

Lemma 7.1 (Central directions). *If λ is central, then*

$$C(\lambda, x) = x, \quad C(x, \lambda) = \lambda$$

whenever the indicated inverses exist. Consequently, after bottom pseudo-roots outside a vertex set V are assigned distinct central scalars, pseudo-roots in directions belonging to V reduce to those for the induced graph $G[V]$, while pseudo-roots in outside directions remain their assigned scalars.

Proof. Both identities follow because $\lambda - x$ commutes with x and with λ . The consequence follows inductively along any sequence of diamond moves constructing the relevant pseudo-root. $\square$

Lemma 7.2. *At a bottom tuple of pairwise distinct central scalars, the Jacobian of $F_{G,n}$ is the identity for every graph G and every matrix size n . Thus $F_{G,n}$ is dominant and generically finite.*

Proof. Every neighborhood pseudo-root is obtained by repeatedly applying C , and at each application Lemma 3.1 preserves the perturbation in the direction of the second argument. Therefore the i th seed has first-order term equal to the perturbation of the i th bottom pseudo-root and no other first-order term. $\square$

Theorem 7.3. *If G contains an induced P_4 , then its neighborhood seed map F_G has no free rational inverse. More precisely,*

$$\deg(F_{G,n}) \geq 3 \quad (n \geq 2).$$

Proof. Let $V = \{v_1, v_2, v_3, v_4\}$ induce the path $v_1 - v_2 - v_3 - v_4$. Assign the scalar centers $0, 1, 2, 3$ to these four bottom directions and assign pairwise distinct rational scalar centers, different from $0, 1, 2, 3$, to all remaining directions.

For 2×2 matrices, place the formal preimage (10) in the four directions in V and keep every outside input equal to its scalar center. Lemma 7.1 shows that its image under $F_{G,2}$ is the curve (9) in the four directions in V and is constant in every outside direction. All defining denominators have invertible scalar terms at $t = 0$.

The field of entries of this branch contains $z(t)$ and therefore has degree at least three over $\mathbb{Q}(t)$. Lemma 7.2 gives Jacobian determinant $1 + O(t)$ on the original branch, and its field conjugates remain nonzero. As in Proposition 6.1, there are at least three distinct simple points in one fiber, so $\deg(F_{G,2}) \geq 3$. Direct-sum amplification with the scalar centers gives the same conclusion for every $n > 2$.

If F_G had a free rational inverse, Lemma 3.4 would make some $F_{G,n}$ birational, contradicting the degree bound. $\square$

Since a graph is a cograph exactly when it has no induced P_4 , combining Theorem 7.3 with the cograph diamond-generation theorem [8, Theorem 1.1(ii)] gives the full rational classification.

Corollary 7.4 (Rational cograph classification). *For every finite graph G over a characteristic-zero base field,*

$$Y_G \text{ rationally generates all Boolean-lattice pseudo-roots} \iff G \text{ is a cograph.}$$

Proof. If G is a cograph, the diamond-generation theorem recovers every pseudo-root from Y_G , and diamond moves are rational operations. If G is not a cograph, it contains an induced P_4 , so Theorem 7.3 rules out rational recovery even of all bottom pseudo-roots. $\square$

The matrix-degree proof of Theorem 7.3 is essential here. Merely substituting central scalars into a hypothetical free rational inverse would not be sufficient: its denominators could vanish identically on the specialized locus. By contrast, the explicit formal branch is defined there, and its simple conjugate points force a degree bound on the unspecialized generic matrix map.

A Reproducibility of the exact calculation

The ancillary script `p4_matrix_degree_certificate.py` was verified with Python 3.12.13 and SymPy 1.14.0 [7]. A matching environment can be prepared with

```
python3 -m pip install sympy==1.14.0
```

and the certificate is run with

```
python3 p4_matrix_degree_certificate.py
```

It checks the Pauli multiplication rules and reversal action, substitutes the formal jet (16) into all four seed equations, verifies (19), constructs the two branch factors S, N , checks their degrees and the resultant identity (23), and verifies the primitive irreducible specialization used in Lemma 5.2. Every failed check raises an explicit exception; the script does not rely on Python assertions, which can be disabled by optimization flags.

References

- [1] P. M. Cohn, *Free Rings and Their Relations*, 2nd ed., London Mathematical Society Monographs 19, Academic Press, London, 1985.
- [2] D. G. Corneil, H. Lerchs, and L. Stewart Burlingham, Complement reducible graphs, *Discrete Appl. Math.* **3** (1981), no. 3, 163–174, doi:10.1016/0166-218X(81)90013-5.
- [3] I. Gelfand, V. Retakh, and R. L. Wilson, Quadratic linear algebras associated with factorizations of noncommutative polynomials and noncommutative differential polynomials, *Selecta Math. (N.S.)* **7** (2001), no. 4, 493–523, doi:10.1007/s00029-001-8096-5.
- [4] I. Gelfand, S. Gelfand, V. Retakh, and R. L. Wilson, Factorizations of polynomials over noncommutative algebras and sufficient sets of edges in directed graphs, *Lett. Math. Phys.* **74** (2005), 153–167, doi:10.1007/s11005-005-0024-8.
- [5] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, New York, 1977, doi:10.1007/978-1-4757-3849-0.

- [6] D. S. Kaliuzhnyi-Verbovetskyi and V. Vinnikov, *Foundations of Free Noncommutative Function Theory*, Mathematical Surveys and Monographs 199, American Mathematical Society, Providence, RI, 2014, doi:10.1090/surv/199.
- [7] A. Meurer et al., SymPy: symbolic computing in Python, *PeerJ Comput. Sci.* **3** (2017), e103, doi:10.7717/peerj-cs.103.
- [8] M. Ramani, Cographs and minimum diamond-generating edge sets in Boolean lattices, arXiv:2607.19048 [math.CO] (2026).
- [9] V. Retakh and M. Saks, On the rational relationships among pseudo-roots of a non-commutative polynomial, *J. Pure Appl. Algebra* **225** (2021), no. 6, Paper No. 106581, doi:10.1016/j.jpaa.2020.106581.